Parametric DPMs

C.Platero

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1 Methods

1.1 Parametric DPM

We denote the biomarker measurement k for individual i at time j as y_{ijk}, where $i = 1, \ldots, n$, $k = 1, \ldots, p$ and $j = 1, \ldots, q_{ik}$. The model can be written as:

$$y_{ijk} = f(\boldsymbol{z}_i, t_{i,j}, \boldsymbol{\Theta}_k) + e_{ijk},$$

where f corresponds to the model, z_i are variables associated to the i-th subject, $t_{i,j}$ is the age of subject i in visit j, Θ_k are the average trajectory parameters, and e_{ijk} is an error term.

In general, the aim is to identify the parameters that most accurately represent the observations, such as those that optimize the likelihood [1, 2] or optimization the robust nonlinear regression [3].

The precision of DPM estimates will increase as patient follow-up is prolonged and more marker data are collected. With a greater number of measurements over time for each subject, individual patient or short-term trajectories will become more distinctive. In contrast, with few visits, patient marker trajectories tend to follow the pattern of the general population or long-term trajectories.

1.2 RPDPM

The variable $s_{i,j}$ transforms the patient's age $t_{i,j}$ into a timescale that defines the progression of the disease, where α_i and β_i represent the temporal scalar parameters for the i-th patient:

$$s_{i,j} = \alpha_i \cdot t_{i,j} + \beta_i.$$

The geometric model for the trajectories is defined by:

$$f(s; \boldsymbol{\Theta}_k) = (a_k - d_k) g\left(s; \boldsymbol{\Theta}_k\right) + d_k$$

where a_k and d_k are normalization scalar parameters for each k-marker, and $g(s; \boldsymbol{\Theta}_k)$ represents a logistic function and $\boldsymbol{\Theta}$ is a vector parameter for each k-marker. Finally, the multiobjective optimization for robust nonlinear regression is defined as:

$$\left\{\hat{\boldsymbol{\alpha}}, \hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\Theta}}\right\} = \min_{i,j,k} \sum_{i,j,k} \omega_i \rho \left(\frac{y_{ijk} - f(z_i, t_{i,j}, \Theta_k)}{\sigma_k} \right)$$

where $\rho(\cdot)$ is a maximum likelihood-type function and $\omega_i = 1/N_i$ is a weighting factor for normalizing the objective function with the number of available points per subject (N_i).

1.3 Leaspy

It combines the concept of time-warp with translations of curves. The transformation of model to the subject's data consist in computing (1) a time-shift, τ_i, which translates the model along the time axis to accommodate for changes in age at disease onset, (2) an acceleration factor α_i, which scales the time interval to accommodate for differences in speed of progression, and (3) inter-marker spacings $\boldsymbol{w}_{ik}$, which translate each biomarker differently to accommodate for differences in

the timing and ordering among biomarkers. The first two parameters define a time-warp function that maps the actual age of the subject to his Alzheimer Age. This function therefore changes the dynamics of progression but not the trajectory. In turn, inter-marker spacings shift the patient trajectory to account for phenotypic differences across subjects [4].

The long-term disease progression corresponds to the geodesic $\gamma 0$, parametrized by $\boldsymbol{p}_0, t_0$ and $\boldsymbol{v}_0$. We use normalized scores on marker values ensuring that geodesics take the form of a logistic curve for each coordinate, so with $p_k = \gamma_k(t_0)$ and $v_k = \dot{\gamma}_k(t_0)$ the multivariate model writes :

$$\gamma_k(t) = \left(1 + \left(\left(\frac{1}{p_k} - 1\right) \cdot exp\left(-\frac{u_k \cdot (t - t_0)}{p_k \cdot (1 - p_k)}\right)\right)\right)^{-1}$$

On the other hand, the function $f(\boldsymbol{z}_i, t_{i,j}, \boldsymbol{\Theta}_k)$ corresponds to the geometrical description of the individual trajectory on the Riemannian manifold, which produces the spatio-temporal transformation from the long-term into the i-short-term trajectory:

$$f(\boldsymbol{z}_i, t_{i,j}, \boldsymbol{\Theta}_k) = \left(1 + \left(\left(\frac{1}{p_k} - 1\right) \cdot exp\left(-\frac{w_{i,k} + u_k \cdot \alpha_i \cdot (t_{i,j} - \tau_i - t_0)}{p_k \cdot (1 - p_k)}\right)\right)\right)^{-1}$$

In order to estimate the diagnosis at each patient consultation, a linear transformation is also used that relates the patient's age to the time of disease progression:

$$s_{i,j} = \alpha_i \cdot t_{i,j} - \alpha \cdot (\tau_i + t_0)$$

1.4 GRACE

The proposal of Donohue et al. [3] is expressed as:

$$f(\boldsymbol{z}_i, t_{i,j}, \boldsymbol{\Theta}_k) = g_k(t^c_{ijk} + \delta_i) + x'_{t^c_{ijk}}\beta_k + \alpha_{0ik} + \alpha_{1ik}t^c_{ijk}$$

where g_k is a continuously differentiable monotone function and δ_i is the unknown subject-specific time shift, which follows a normal distribution with mean zero and variance σ^2_δ. Short-term observation time is represented by t^c_{ijk}, which indicates the centered years in relation to the temporal evolution of the visits, $t^c_{ij} = t_{ij} - (t_{i_1} + t_{i_{end}})/2$, where t_{ij} is the visit time and i_1 and i_{end} represented the first and last visit index of $i-$subject. $x'_{t^c_{ijk}}$ is the row vector for the fixed effects (including variables such as age and scan time), and β_k are the fixed effects coefficients. The long-term progression time is computed from $t^c_{ijk} + \delta_i$. The parameters α_{0ik} and α_{1ik} are the subject- and outcome-specific random intercept and slope. These values reflect how the subset of regression parameters for the ith subject deviates from those of the population.

2 DPM's performances

Assessing the performance of DPM is essential. This evaluation can be carried out through two types of measures: a) Measuring the accuracy in estimating the temporal trajectories of patients' markers. b) Evaluating patients' diagnosis over time, including estimates of remaining time for key clinical events, such as the age at which cognitive decline or dementia begins.

2.1 Evaluations of estimated trajectories

The Mean Absolute Error (MAE) is a performance measure for models that is less sensitive to outliers (Chai and Draxler, 2014), calculated based on the absolute differences between actual and estimated values:

$$MAE_k = \frac{1}{N_k} \sum \left| y_{y,j,k} - \hat{y}_(i,j,k) \right|$$

N_k represents the total number of measurements for the k-th marker across all subjects and visits. The measure and estimated value of this marker for the i-th subject at the j-th visit are denoted by $y_{i,j,k}$ and $\hat{y}(i,j,k)$ respectively. To enable cross-variable comparisons, the MAEs are normalized as NMAE.

2.2 Clinical status classification

Given a disease progression score for the j-th visit of the i-th subject, $s_{i,j}$, the clinical status is classified based on a Bayesian classifier that uses the kernel density estimation (KDE) (Parzen, 1962)-based fitted likelihoods as:

$$p(s|c) = \frac{1}{n \cdot h} \sum_{t=1}^{n} K\left(\frac{s - s_t}{h}\right)$$

where $K(\cdot)$ is the kernel, $h \geq 0$ is a smoothing parameter and c denotes a clinical group. The predicted class for observation $s_{i,j}$ is c_r, which maximizes the posterior probability:

$$r = \arg\max_r \left(p(c_r) \cdot p(s_{i,j}|c_r) \right).$$

2.3 Measures regarding the timeframes of conversion toward cognitive decline

Quality measures were proposed for DPMs associated with a specific clinical event: in this case, the onset of cognitive decline in a patient. Firstly, the estimated age at which the subject initiates this decline was calculated according to a proposed model. This date was determined as the average time between the first visit diagnosed with MCI and the preceding visit where the patient was considered to have normal cognitive function. Additionally, the patient's final visit in the longitudinal study must confirm the patient's cognitive decline or dementia status. Based on the proposed date for this clinical event in the i-patient, denoted as $Age_{i_{onset}}$, the following measures are suggested:

1. For sCU subjects, the age at the event must be greater than the patient's age at the last visit:

$$SCORE_1 = \frac{\#\left\{i|(i \in sCU) \cap ((Age_{i_{end}} < Age_{i_{onset}})\right\}}{\#sCU}$$

For sMCI subjects, the age at event must be the same as the age at baseline:

$$SCORE_2 = \frac{\#\left\{i|(i \in sMCI) \cap ((Age_{i_1} < Age_{i_{onset}})\right\}}{\#sMCI}$$

Regarding subjects progressing from CU to MCI, we measured: a) the percentage of preclinical CU cases detected by the DPMs:

$$SCORE_3 = \frac{\#\left\{i|(i \in pCU) \cap ((Age_{i_1} < Age_{i_{onset}})\right\}}{\#pCU}$$

$$SCORE_4 = \frac{\#\left\{i|(i \in pCU) \cap ((Age_{i_{end}} \geq Age_{i_{onset}})\right\}}{\#pCU}$$

2. The correlation between age at the event and its estimation by the model.

3. The correlation between the years remaining for the individual to reach cognitive decline according to clinical diagnosis and the model's estimation from the visit's date at baseline.

2.4 Data partitions and boostraping

To assess the algorithms, the dataset is divided into two non-overlapping sets for training and testing. Furthermore, bootstrapping is employed in the experiments to evaluate the DPM's performances. Within each bootstrap iteration, a subset of the training subjects is randomly sampled with replacement, taking into account their clinical grouping, which ensures that each bootstrap sample includes data representing all diagnostic statuses.

References

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